

Notes on Linear Algebra

DSBA Mathematics Refresher 2025

Abstract

1 Linear Independence Property

Given a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V , the set is said to be **linearly independent** if the only solution to the equation

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}$$

is

$$c_1 = c_2 = \dots = c_n = 0.$$

In other words, none of the vectors in the set can be written as a linear combination of the others.

Example: Consider the vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 2 \\ 4 \\ -1 \\ 2 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} -2 \\ 5 \\ 0 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_3 = \begin{pmatrix} 1 \\ 0 \\ 5 \\ 3 \\ -4 \end{pmatrix}$ in

$\mathbb{R}^5$. These vectors are linearly independent because the only solution to

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{0}$$

is $c_1 = c_2 = c_3 = 0$. Indeed, writing the equation component by component gives the system

$$\begin{cases} 3c_1 - 2c_2 + c_3 &= 0 \\ 2c_1 + 5c_2 &= 0 \\ 4c_1 + 5c_3 &= 0 \\ -c_1 + c_2 + 3c_3 &= 0 \\ 2c_1 + c_2 - 4c_3 &= 0 \end{cases}$$

The second line gives $c_2 = -\frac{2}{5}c_1$ and the third gives $c_3 = -\frac{4}{5}c_1$; substituting both into the first line:

$$3c_1 + \frac{4}{5}c_1 - \frac{4}{5}c_1 = 3c_1 = 0 \quad \implies \quad c_1 = 0, \quad \text{and hence} \quad c_2 = c_3 = 0$$

Counter-example: The vectors $\mathbf{w}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $\mathbf{w}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{w}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ in $\mathbb{R}^3$ are **not** linearly independent, because

$$\mathbf{w}_1 + \mathbf{w}_2 - \mathbf{w}_3 = \mathbf{0}$$

is a non-trivial linear combination giving $\mathbf{0}$ (here $c_1 = 1$, $c_2 = 1$, $c_3 = -1$). Equivalently, $\mathbf{w}_3 = \mathbf{w}_1 + \mathbf{w}_2$: the third vector carries no information that the first two do not already carry.

2 Spanning Property

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V is said to **span** V if every vector in V can be expressed as a linear combination of these vectors. Mathematically, the set spans V if for every vector $\mathbf{v} \in V$, there exist scalars $c_1, c_2, \dots, c_n$ such that

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n.$$

Example: In $\mathbb{R}^2$, the vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ and $\mathbf{v}_3 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ span $\mathbb{R}^2$ because any vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$ can be written as

$$\mathbf{v} = \frac{2x+y}{3} \mathbf{v}_1 + \frac{y-x}{3} \mathbf{v}_2 + 0 \cdot \mathbf{v}_3.$$

Indeed,

$$\frac{2x+y}{3} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{y-x}{3} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{2x+y-y+x}{3} \\ \frac{2x+y+2y-2x}{3} \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}.$$

Note that such a decomposition is **not unique** here: the three vectors are redundant (one checks that $\mathbf{v}_3 = \frac{1}{3}\mathbf{v}_1 - \frac{2}{3}\mathbf{v}_2$), and we could equally have written $\mathbf{v} = \frac{x+y}{2}\mathbf{v}_1 + 0 \cdot \mathbf{v}_2 + \frac{x-y}{2}\mathbf{v}_3$. Spanning only requires that *at least one* decomposition exists.

Counter-example: In $\mathbb{R}^3$, the vectors $\mathbf{w}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $\mathbf{w}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ do **not** span $\mathbb{R}^3$: every linear combination $c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 = \begin{pmatrix} c_1 \\ c_2 \\ 0 \end{pmatrix}$ has a third component equal

to 0, so for instance $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ cannot be reached. They span the plane $\{z = 0\}$, which is a strict subspace of $\mathbb{R}^3$.

3 Basis

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in a vector space V is called a **basis** for V if:

- The set is linearly independent.
- The set spans the vector space V .

The number of vectors in a basis is called the **dimension** of the vector space.

Example: The vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ form a basis for $\mathbb{R}^2$. The dimension of $\mathbb{R}^2$ is 2.

Example (a basis is not unique): The vectors $\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{u}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ also form a basis for $\mathbb{R}^2$:

- They are linearly independent: $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 = \mathbf{0}$ gives $c_1 + c_2 = 0$ and $c_1 - c_2 = 0$, so $c_1 = c_2 = 0$.
- They span $\mathbb{R}^2$: any $\begin{pmatrix} x \\ y \end{pmatrix} = \frac{x+y}{2}\mathbf{u}_1 + \frac{x-y}{2}\mathbf{u}_2$.

A vector space has many different bases, but they all have the *same number of vectors*, which is why the dimension is well defined.

Counter-examples:

- The three vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ of the previous section span $\mathbb{R}^2$, but they are *not* a basis: 3 vectors in a 2-dimensional space can never be linearly independent.
- The two vectors $\mathbf{w}_1, \mathbf{w}_2$ of the previous section are linearly independent in $\mathbb{R}^3$, but they are *not* a basis of $\mathbb{R}^3$ either: they do not span it.

In general, in a space of dimension n , a basis has *exactly* n vectors: fewer cannot span, more cannot be independent.

4 (Reduced) Row Echelon Form

A matrix is in **row echelon form** if it satisfies the following conditions:

1. All non-zero rows are above any rows of all zeros.
2. The leading entry of each non-zero row after the first occurs to the right of the leading entry of the previous row.
3. The leading entry in any non-zero row is 1 (this condition is for the **reduced** row echelon form).
4. The leading 1 is the only non-zero entry in its column (for **reduced** row echelon form).

Example: The matrix

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$$

is in row echelon form. It is *not* in **reduced** row echelon form, because the leading 1's are not the only non-zero entries of their columns. Clearing the entries above the pivots, with $L_2 \leftarrow L_2 - 4L_3$, then $L_1 \leftarrow L_1 - 3L_3$ and finally $L_1 \leftarrow L_1 - 2L_2$, gives the reduced row echelon form:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Example (a full reduction): Let us reduce

$$M = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 0 \\ 3 & 6 & 2 & 1 \end{pmatrix}$$

First, use the pivot in position (1, 1) to clear the entries below it, with $L_2 \leftarrow L_2 - 2L_1$ and $L_3 \leftarrow L_3 - 3L_1$:

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -6 \\ 0 & 0 & 5 & -8 \end{pmatrix}$$

The second column has no pivot (it is a **free** column), so we move to the third one. With $L_2 \leftarrow \frac{1}{3}L_2$, then $L_3 \leftarrow L_3 - 5L_2$ and $L_3 \leftarrow \frac{1}{2}L_3$:

$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

This is a row echelon form. To make it *reduced*, we clear the entries above each pivot, with $L_2 \leftarrow L_2 + 2L_3$, then $L_1 \leftarrow L_1 + L_2$ and $L_1 \leftarrow L_1 - 3L_3$:

$$\begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

which is the reduced row echelon form of M (with pivots in columns 1, 3 and 4).

Example (solving a system): This is what row reduction is mostly used for. To solve

$$\begin{cases} x + 2y + z &= 8 \\ 2x + y - z &= 1 \\ 3x - y + 2z &= 7 \end{cases}$$

we write the **augmented matrix** of the system, and reduce it. With $L_2 \leftarrow L_2 - 2L_1$ and $L_3 \leftarrow L_3 - 3L_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & 1 & -1 & 1 \\ 3 & -1 & 2 & 7 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -3 & -3 & -15 \\ 0 & -7 & -1 & -17 \end{array} \right)$$

With $L_2 \leftarrow -\frac{1}{3}L_2$, then $L_3 \leftarrow L_3 + 7L_2$ and $L_3 \leftarrow \frac{1}{6}L_3$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 1 & 1 & 5 \\ 0 & -7 & -1 & -17 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 3 \end{array} \right)$$

The last row reads $z = 3$; substituting back into the second row, $y + 3 = 5$, so $y = 2$; and into the first, $x + 4 + 3 = 8$, so $x = 1$. This last stage is called **back-substitution**.

$$x = 1, \quad y = 2, \quad z = 3$$

Check: $1 + 4 + 3 = 8$ ✓, $2 + 2 - 3 = 1$ ✓, $3 - 2 + 6 = 7$. ✓

5 Inverting a Matrix

The **inverse** of a square matrix A is denoted by A^{-1} and is defined as the matrix that satisfies

$$AA^{-1} = A^{-1}A = I,$$

where I is the identity matrix.

Steps to find the inverse of a matrix:

1. Form the augmented matrix $[A|I]$.
2. Use row operations to convert A into the identity matrix.
3. The matrix that results from the identity matrix on the left side is A^{-1} on the right side.

Example: For the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$

the augmented matrix is

$$[A|I] = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right)$$

after reducing to row echelon form:

$$[I|A^{-1}] = \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & 1.5 & -0.5 \end{array} \right)$$

hence, the inverse is

$$A^{-1} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}$$

In detail, the row operations are $L_2 \leftarrow L_2 - 3L_1$, then $L_2 \leftarrow -\frac{1}{2}L_2$, and finally $L_1 \leftarrow L_1 - 2L_2$:

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right) \longrightarrow \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 1.5 & -0.5 \end{array} \right) \longrightarrow \left(\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & 1.5 & -0.5 \end{array} \right)$$

It is always worth checking the result:

$$AA^{-1} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix} = \begin{pmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \quad \checkmark$$

Example (3×3): For

$$N = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix},$$

we start from $[N|I]$ and apply $L_3 \leftarrow L_3 - 5L_1$, then $L_3 \leftarrow L_3 + 4L_2$:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 5 & 6 & 0 & 0 & 0 & 1 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & -4 & -15 & -5 & 0 & 1 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array} \right)$$

The left block is now in row echelon form; we finish with $L_1 \leftarrow L_1 - 2L_2$, then $L_2 \leftarrow L_2 - 4L_3$ and $L_1 \leftarrow L_1 + 5L_3$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -5 & 1 & -2 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -24 & 18 & 5 \\ 0 & 1 & 0 & 20 & -15 & -4 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array} \right)$$

hence

$$N^{-1} = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix}$$

(one checks that $NN^{-1} = I$).

6 Determinant

The **determinant** of a square matrix A , denoted by $\det(A)$, is a scalar value that can be computed from the elements of the matrix. The determinant has important properties and applications, including:

- A matrix is invertible if and only if its determinant is non-zero.
- The determinant of a product of matrices is the product of their determinants: $\det(AB) = \det(A)\det(B)$.

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is calculated as:

$$\det(A) = ad - bc.$$

For a 3×3 matrix $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, the determinant is calculated as:

$$\det(A) = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}.$$

Expanding the 2×2 determinants (minors):

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

For higher dimension matrices, ask a computer (it's outside the scope of this course).

Example: For the matrix

$$B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix},$$

$\det B = 0$, hence, the matrix is not invertible (you can try to invert it, you will encounter a problem).

While for the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$

$\det A = -2 \neq 0$, hence, the matrix is invertible (and we found the inverse in the previous section).

Example (3×3): For the matrix

$$N = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$$

of the previous section, expanding along the first row:

$$\det(N) = 1 \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} - 2 \begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} + 3 \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix}$$

$$= 1 \times (0 - 24) - 2 \times (0 - 20) + 3 \times (0 - 5) = -24 + 40 - 15 = 1 \neq 0$$

so N is invertible, which is consistent with the inverse we computed in the previous section.

Remark: the determinant is often the quickest way to know *whether* a matrix can be inverted, but it does not give the inverse. When a matrix is *not* invertible, it means its columns are not linearly independent: for $B = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ above, the second column is twice the first.

7 Eigenvalues

Let A be an $n \times n$ matrix. A scalar λ is called an **eigenvalue** of A if there exists a non-zero vector $\mathbf{v} \in \mathbb{R}^n$ (or $\mathbb{C}^n$) such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The vector $\mathbf{v}$ is referred to as the corresponding **eigenvector** of A associated with λ .

Finding Eigenvalues To find the eigenvalues of a matrix A , we solve the characteristic equation:

$$\det(A - \lambda I) = 0,$$

where I is the identity matrix of the same size as A , and $\det(\cdot)$ denotes the determinant. The solutions λ are the eigenvalues of A .

Example: Consider the matrix

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

The characteristic equation is given by:

$$\begin{aligned} \det(A - \lambda I) &= \det\left(\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) \\ &= \det\left(\begin{pmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}\right) = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = 0. \end{aligned}$$

The eigenvalues are the roots of this quadratic equation, $\lambda_1 = 5$ and $\lambda_2 = 2$.

Example (triangular matrix): For a triangular matrix, the eigenvalues can be read directly on the diagonal. Indeed, for

$$T = \begin{pmatrix} 2 & 7 & -1 \\ 0 & 5 & 3 \\ 0 & 0 & 1 \end{pmatrix},$$

the matrix $T - \lambda I$ is still triangular, and the determinant of a triangular matrix is the product of its diagonal entries:

$$\det(T - \lambda I) = (2 - \lambda)(5 - \lambda)(1 - \lambda) = 0 \quad \implies \quad \lambda = 2, \lambda = 5 \text{ or } \lambda = 1$$

Example (complex eigenvalues): Eigenvalues are not always real. For the rotation matrix

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

the characteristic equation is $\det(R - \lambda I) = \lambda^2 + 1 = 0$, whose solutions are $\lambda = i$ and $\lambda = -i$. This makes sense geometrically: R rotates every vector of $\mathbb{R}^2$ by 90° , so no non-zero *real* vector keeps its direction.

Example (repeated eigenvalue): For

$$S = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix},$$

the characteristic equation is $(3 - \lambda)^2 = 0$, so $\lambda = 3$ is a *single* eigenvalue, counted with multiplicity 2. We will see in the section on diagonalization why this case deserves attention.

8 Eigenvectors

Given an eigenvalue λ of a matrix A , the corresponding **eigenvector** $\mathbf{v}$ is any non-zero vector that satisfies the equation:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

To find the eigenvectors associated with λ , we solve the system:

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

This is a system of linear equations.

Example: For the matrix A from the previous example and $\lambda_1 = 5$, we solve:

$$(A - 5I)\mathbf{v} = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0}.$$

One solution to this system is $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ ¹.

Similarly, for $\lambda_2 = 2$, we solve:

$$(A - 2I)\mathbf{v} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0},$$

which gives $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ ².

Example (in practice): Rather than inverting anything, it is often quickest to write $A\mathbf{v} = \lambda\mathbf{v}$ line by line. For $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ and $\lambda_1 = 5$, with $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$:

$$\begin{cases} 4x + y = 5x \\ 2x + 3y = 5y \end{cases} \iff \begin{cases} y = x \\ 2x = 2y \end{cases} \iff y = x$$

¹Any vector of the form $\mathbf{v} = \begin{pmatrix} \nu \\ \nu \end{pmatrix}$, $\nu \in \mathbb{R}^*$ would work, it is common practice to set $\nu = 1$.

²Again, any vector of the form $\mathbf{v} = \begin{pmatrix} -\nu \\ 2\nu \end{pmatrix}$, $\nu \in \mathbb{R}^*$ would work, it is common practice to set $\nu = 1$.

The two equations say the same thing, which is expected: if they did not, the only solution would be $\mathbf{v} = \mathbf{0}$, and λ would not be an eigenvalue. Setting $x = 1$ gives back $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Example (repeated eigenvalue): For the matrix $S = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$ above, with its single eigenvalue $\lambda = 3$:

$$(S - 3I)\mathbf{v} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ 0 \end{pmatrix} = \mathbf{0} \quad \implies \quad y = 0$$

so every eigenvector is a multiple of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Even though the eigenvalue is repeated *twice*, we only get *one* independent eigenvector: this is exactly the situation that prevents diagonalization.

9 Diagonalization

A square matrix A is said to be **diagonalizable** if there exists an invertible matrix P and a diagonal matrix D such that:

$$A = PDP^{-1}.$$

When this is possible, the diagonal elements of D are the eigenvalues of A , and the columns of P are the corresponding eigenvectors.

Procedure for Diagonalization

1. Find the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of A .
2. For each eigenvalue λ_i , find the corresponding eigenvector $\mathbf{v}_i$.
3. Form the matrix P using the eigenvectors as columns: $P = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_n)$.
4. The matrix D is a diagonal matrix with eigenvalues on the diagonal:

$$D = \begin{pmatrix} \lambda_1 & & 0 & 0 & 0 \\ & \lambda_2 & & \ddots & 0 \\ 0 & & \ddots & & 0 \\ 0 & \ddots & & \ddots & \\ 0 & 0 & 0 & & \lambda_n \end{pmatrix}.$$

5. Verify that $A = PDP^{-1}$.

Example: Consider the matrix $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$. From our earlier work, the eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = 2$, with corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

Thus, we can form:

$$P = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}.$$

Finally, verify that:

$$P^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}, \quad A = PDP^{-1} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

In detail:

$$PDP^{-1} = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 5 & -2 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 12 & 3 \\ 6 & 9 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \quad \checkmark$$

Counter-example (not every matrix is diagonalizable): Consider again $S = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$. Its only eigenvalue is $\lambda = 3$, and all its eigenvectors are multiples of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. We can therefore not build a 2×2 matrix P with two *linearly independent* eigenvectors as columns: any such P would have two proportional columns, hence $\det(P) = 0$, and P would not be invertible. The matrix S is **not diagonalizable**.

In general, an $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors. Two useful sufficient conditions:

- if the matrix has n *distinct* eigenvalues, it is diagonalizable (as in the example above);
- if the matrix is *symmetric* (with real entries), it is diagonalizable, and P can even be chosen orthogonal. This is the case that matters for Principal Component Analysis.

Why bother? Because a diagonal matrix is much easier to compute with. For instance, powers of A become trivial: since

$$A^k = \underbrace{PDP^{-1} \cdot PDP^{-1} \cdots PDP^{-1}}_{k \text{ times}} = PD^kP^{-1}$$

(the inner $P^{-1}P = I$ cancel out), and D^k is simply the diagonal matrix of the λ_i^k , we get for our example

$$A^k = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5^k & 0 \\ 0 & 2^k \end{pmatrix} \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$$

without ever multiplying A by itself. This is what makes diagonalization useful for iterated processes (Markov chains, dynamical systems, ...).

10 Norms

In vector spaces, a **norm** is a function that assigns a non-negative length or size to vectors. Norms are widely used to measure the magnitude of vectors. The most common norms are the L_1 , L_2 , and L_p norms.

L_1 Norm (Manhattan Norm) The L_1 norm of a vector $\mathbf{v} = (v_1, v_2, \dots, v_n)$ in $\mathbb{R}^n$ is defined as the sum of the absolute values of its components:

$$\|\mathbf{v}\|_1 = |v_1| + |v_2| + \dots + |v_n| = \sum_{i=1}^n |v_i|.$$

This norm is also known as the **Manhattan norm** or **taxicab norm**, as it represents the distance between two points in a grid-based path, like the blocks in a city.

Example: For the vector $\mathbf{v} = \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$ in $\mathbb{R}^3$, the L_1 norm is:

$$\|\mathbf{v}\|_1 = |3| + |-4| + |1| = 3 + 4 + 1 = 8.$$

L_2 Norm (Euclidean Norm) The L_2 norm of a vector $\mathbf{v} = (v_1, v_2, \dots, v_n)$ is the square root of the sum of the squares of its components:

$$\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2} = \sqrt{\sum_{i=1}^n v_i^2}.$$

This norm is commonly known as the **Euclidean norm** because it represents the usual notion of distance in Euclidean space.

Example: For the vector $\mathbf{v} = \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$ in $\mathbb{R}^3$, the L_2 norm is:

$$\|\mathbf{v}\|_2 = \sqrt{3^2 + (-4)^2 + 1^2} = \sqrt{9 + 16 + 1} = \sqrt{26}.$$

L_p Norm The L_p norm generalizes the L_1 and L_2 norms and is defined for any real number $p \geq 1$. The L_p norm of a vector $\mathbf{v} = (v_1, v_2, \dots, v_n)$ is given by:

$$\|\mathbf{v}\|_p = (|v_1|^p + |v_2|^p + \dots + |v_n|^p)^{\frac{1}{p}} = \left(\sum_{i=1}^n |v_i|^p \right)^{\frac{1}{p}}.$$

Different values of p result in different norms. The L_2 norm is a special case where $p = 2$, and the L_1 norm is a special case where $p = 1$.

Example: For the vector $\mathbf{v} = \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$ in $\mathbb{R}^3$, the L_p norm with $p = 3$ is:

$$\|\mathbf{v}\|_3 = (|3|^3 + |-4|^3 + |1|^3)^{\frac{1}{3}} = (27 + 64 + 1)^{\frac{1}{3}} = \sqrt[3]{92}.$$

Properties of Norms To be a norm, a function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ must verify:

1. **Non-negativity:** $\|\mathbf{v}\| \geq 0$ for all vectors $\mathbf{v}$, and $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.
2. **Scalar Multiplication:** For any scalar α and vector $\mathbf{v}$, $\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\|$.
3. **Triangle Inequality:** For any vectors $\mathbf{v}$ and $\mathbf{w}$, $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$.

Example: Take $\mathbf{v} = \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}$, so that $\mathbf{v} + \mathbf{w} = \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$.

- *Scalar multiplication:* $2\mathbf{v} = \begin{pmatrix} 6 \\ -8 \\ 2 \end{pmatrix}$, and $\|2\mathbf{v}\|_1 = 6 + 8 + 2 = 16 = 2 \times 8 = 2\|\mathbf{v}\|_1$. ✓

- *Triangle inequality, for the L_1 norm:*

$$\|\mathbf{v} + \mathbf{w}\|_1 = 2 + 2 + 3 = 7 \leq \|\mathbf{v}\|_1 + \|\mathbf{w}\|_1 = 8 + 5 = 13 \quad \checkmark$$

- *Triangle inequality, for the L_2 norm:*

$$\|\mathbf{v} + \mathbf{w}\|_2 = \sqrt{4 + 4 + 9} = \sqrt{17} \approx 4.12 \leq \|\mathbf{v}\|_2 + \|\mathbf{w}\|_2 = \sqrt{26} + 3 \approx 8.10 \quad \checkmark$$

The inequality is an equality only when $\mathbf{v}$ and $\mathbf{w}$ point in the same direction; here they do not, so there is some slack.

Why it matters in practice: when the vector measures an *error*, the choice of norm changes what the model is asked to avoid. Compare the two error vectors

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{e}_2 = \begin{pmatrix} 4 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

They have the *same* L_1 norm ($\|\mathbf{e}_1\|_1 = \|\mathbf{e}_2\|_1 = 4$), but different L_2 norms: $\|\mathbf{e}_1\|_2 = 2$ while $\|\mathbf{e}_2\|_2 = 4$. An L_2 loss strongly prefers many small errors over one large one; an L_1 loss is indifferent between the two.

11 The Gram-Schmidt Algorithm

We assume the following:

- The vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ are linearly independent in an inner product space V .
- The space V is equipped with an inner product $\langle \mathbf{u}, \mathbf{v} \rangle$ which is usually the standard dot product in $\mathbb{R}^n$.

Given a set of linearly independent vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, the goal is to find an orthogonal set of vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ such that:

- $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$ for all k ,
- $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for all $i \neq j$ (orthogonality),
- Optionally, we can normalize the vectors so that $\|\mathbf{u}_i\| = 1$, forming an orthonormal set.

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be a set of linearly independent vectors. The Gram-Schmidt process proceeds as follows:

Step 1: Construct the first orthogonal vector

We begin by setting the first vector in the orthogonal set equal to the first vector in the original set:

$$\mathbf{u}_1 = \mathbf{v}_1$$

At this point, $\mathbf{u}_1$ is trivially orthogonal because it's the only vector in the set.

Step 2: Orthogonalize the remaining vectors

For each subsequent vector $\mathbf{v}_k$ (where $k = 2, 3, \dots, n$), we subtract off the projections onto the previously calculated orthogonal vectors $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$.

The orthogonal vector $\mathbf{u}_k$ is computed as:

$$\mathbf{u}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \frac{\langle \mathbf{v}_k, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle} \mathbf{u}_j$$

This formula ensures that $\mathbf{u}_k$ is orthogonal to each of the previous vectors $\mathbf{u}_1, \dots, \mathbf{u}_{k-1}$.

Step 3: Normalize (Optional)

If we want an orthonormal set of vectors, we normalize each $\mathbf{u}_k$ by dividing it by its magnitude:

$$\mathbf{e}_k = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|}$$

The vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ now form an orthonormal basis for the same subspace.

Worked Example

Let us orthogonalize the following linearly independent vectors of $\mathbb{R}^3$:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Step 1. We simply take

$$\mathbf{u}_1 = \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \text{with} \quad \langle \mathbf{u}_1, \mathbf{u}_1 \rangle = 1 + 1 + 0 = 2$$

Step 2 (second vector). Since $\langle \mathbf{v}_2, \mathbf{u}_1 \rangle = 1 \times 1 + 0 \times 1 + 1 \times 0 = 1$,

$$\mathbf{u}_2 = \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix}$$

We check that it worked: $\langle \mathbf{u}_2, \mathbf{u}_1 \rangle = \frac{1}{2} - \frac{1}{2} + 0 = 0$. ✓

Step 2 (third vector). We now subtract *two* projections. With $\langle \mathbf{v}_3, \mathbf{u}_1 \rangle = 0 + 1 + 0 = 1$, $\langle \mathbf{v}_3, \mathbf{u}_2 \rangle = 0 - \frac{1}{2} + 1 = \frac{1}{2}$ and $\langle \mathbf{u}_2, \mathbf{u}_2 \rangle = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$:

$$\begin{aligned} \mathbf{u}_3 &= \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 - \frac{\langle \mathbf{v}_3, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \frac{1/2}{3/2} \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 - \frac{1}{2} - \frac{1}{6} \\ 1 - \frac{1}{2} + \frac{1}{6} \\ 1 - \frac{1}{3} \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{pmatrix} \end{aligned}$$

Again we check: $\langle \mathbf{u}_3, \mathbf{u}_1 \rangle = -\frac{2}{3} + \frac{2}{3} + 0 = 0$ ✓ and $\langle \mathbf{u}_3, \mathbf{u}_2 \rangle = -\frac{1}{3} - \frac{1}{3} + \frac{2}{3} = 0$. ✓

Step 3 (normalization). The norms are

$$\|\mathbf{u}_1\| = \sqrt{2}, \quad \|\mathbf{u}_2\| = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}, \quad \|\mathbf{u}_3\| = \sqrt{\frac{4}{3}} = \frac{2}{\sqrt{3}}$$

so the orthonormal basis is

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{e}_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

(for $\mathbf{e}_2$ and $\mathbf{e}_3$, we first factored out $\frac{1}{2}$ and $\frac{2}{3}$ respectively, which is allowed since scaling a vector does not change its direction).

Remark: the order of the vectors matters. Starting from $\mathbf{v}_2$ instead of $\mathbf{v}_1$ would give a different orthonormal basis, spanning the same space. Only the first vector is kept unchanged (up to its length) by the algorithm.